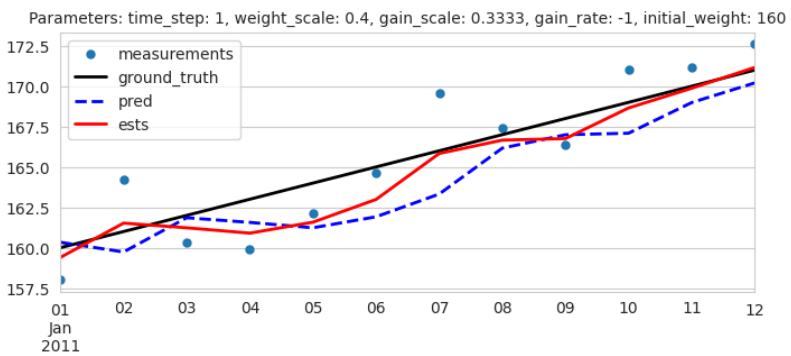
Example of Estimating Body Weight

- Two **problems**
  - The filter needs a correct guess of the weight change rate
  - The rate of change can vary over time
- **Solution**: estimate the rate of change from measurements
  - "Data is better than a guess, even if it's noisy"
  - Refine the estimate of the gain rate:

    new gain = old gain + 0.3 (measurement - prediction) / 1 day



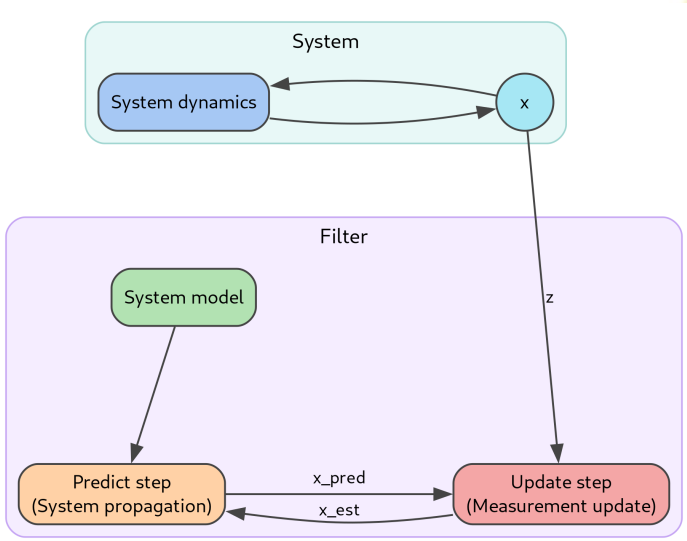
- The "state" is given by `weight` and `gain_rate`
  - You need to predict and update both

g-h Filter

- The previous algorithm is called **g-h filter**
  - g: scaling used to blend predicted state and measurement
  - h: scaling used to update the parameter of the system model based on the measurements
- g-h filters have different values of g and h to achieve different properties
  - E.g., pick g to minimize the transient error when the derivative of the signal has a step (i.e., a discontinuity of the slope)
- Many filters (including Kalman filter) are just generalizations of a g-h filter

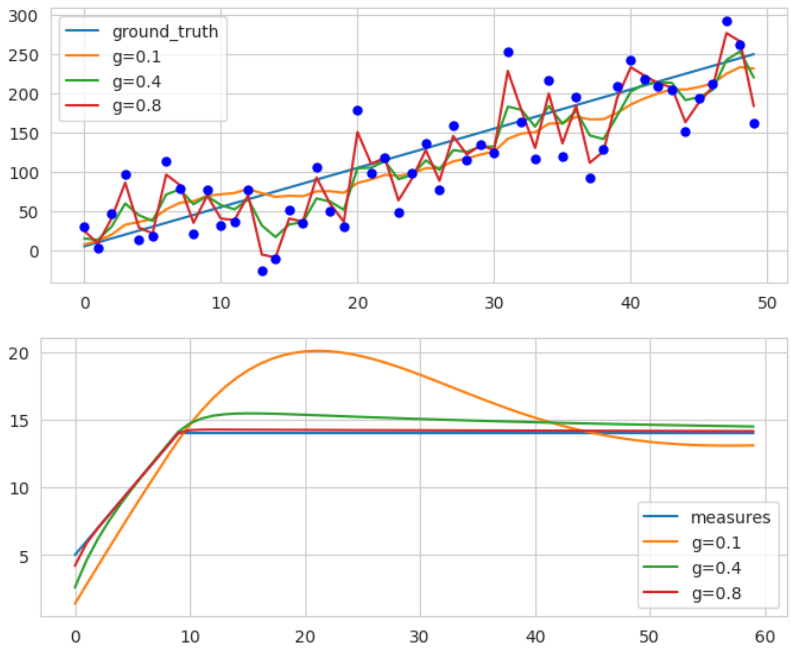
g-h Filter Algorithm: Pseudo-Code

1. **Initialization**
   - Initialize state of the filter
   - Initialize belief in the state
2. **Predict**
   - Use system model to predict state at next time step
   - Adjust belief to account for uncertainty in prediction
3. **Update**
   - Get measurement and associated belief about its accuracy
   - Use as estimate of the next state a point between estimated state and measurement



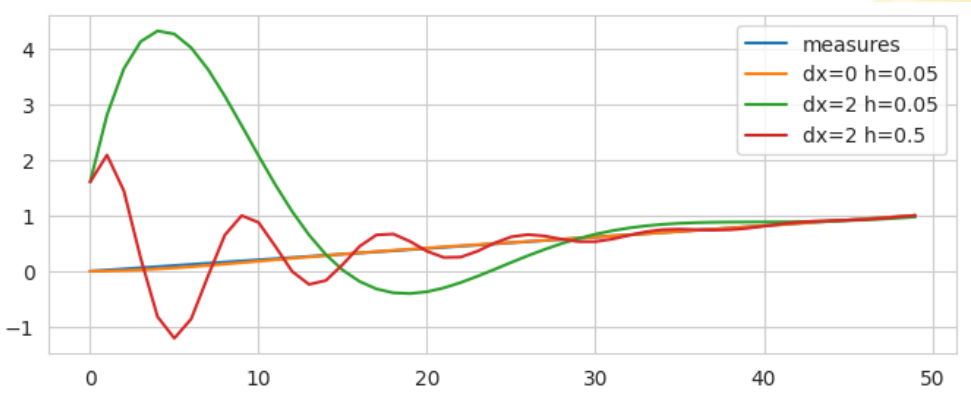
Interpretation of g

- If $g = 0$:
  - Follow system model, ignore measurements
- If g is small:
  - Follow model more than measurements
- If g increases:
  - Follow measurements more, ignore prediction
  - Useful with accurate measurements, inaccurate model
- If $g = 1$:
  - Follow only measurements, ignore model
  - Too large $\rightarrow$ follow measurements, reject no noise



Interpretation of h

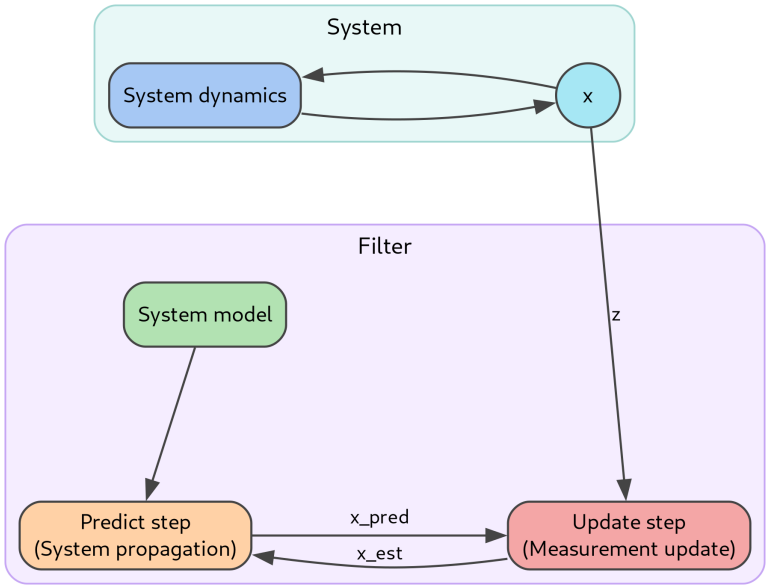
- Estimate model parameters from data:
  - Change of weight
  - Rate of change of measurements
  - Car speed on different terrains



- If $h = 0$:
  - Filter follows previous rate of change
  - Adapts slowly to signal changes
- If $h = 1$:
  - Filter reacts rapidly to significant signal variations
- If h is:
  - Smaller $\rightarrow$ Smoother, slower rate adaptation
  - Larger $\rightarrow$ Faster rate adjustment
- **Note**: Incorrect initial state is similar to a changing state

Control Theory Nomenclature

- State space models developed in control theory → different nomenclature
- **System**: object to estimate/track
- **Filter**: algorithm to estimate system state
- **State of the system** `x`: current values of interest
  - Part of the state might be hidden



- **Measurement** `z`: measured value of system
  - Observable, can be inaccurate
- **State estimate** `x_est`: filter estimate of state
- **System model**: mathematical model of system
  - Typically has error
- **System propagation**: predict step using system model to form new state estimate `x_pred`

- ***One Dimensional Kalman Filters***
- Multivariate Gaussians
- Multivariate Kalman Filters

Updating Belief Using Gaussians

- The **Kalman filter** is based on Bayes theorem and Gaussians
- The Bayes theorem tells that:

$$\text{posterior} = \text{normalized}(\text{prior} \times \text{likelihood})$$

- If the prior and the likelihood are Gaussian the posterior is also Gaussian (conjugate prior)
  - Everything can be represented as Gaussians $\rightarrow$ encode the PDF in terms of mean and std dev
  - Updating belief is equivalent to sum and multiplication of Gaussians
- Algorithm:

```
# Create prior (using current estimate and system model).
prior = predict(x, process_model)

# Create likelihood (using measurement).
likelihood = gaussian(z, sensor_var)

# Update belief using prior and likelihood
posterior = update(prior, likelihood)
```

Sum of Gaussians

- The sum of two independent Gaussians

$$Normal(\mu_1, \sigma_1^2)$$
$$Normal(\mu_2, \sigma_2^2)$$

  is a Gaussian $Normal(\mu, \sigma^2)$ with:

$$\mu = \mu_1 + \mu_2$$
$$\sigma^2 = \sigma_1^2 + \sigma_2^2$$

- The mean is the sum of the mean (by linearity)
- The variance always increases

Product of Gaussians

- The product of two independent Gaussians

$$Normal(\mu_1, \sigma_1^2)$$
$$Normal(\mu_2, \sigma_2^2)$$

is a Gaussian $\mathrm{N}(\mu, \sigma^2)$ with:

$$\mu = \frac{\mu_1\sigma_2^2 + \mu_2\sigma_1^2}{\sigma_1^2 + \sigma_2^2}$$
$$\sigma^2 = \frac{\sigma_1^2\sigma_2^2}{\sigma_1^2 + \sigma_2^2}$$

- **Interpretation:**
  - The variance may be reduced as more information is incorporated
  - If one Gaussian N_1 is much narrower than the other (i.e., one measure is more accurate), the result is pushed towards N_1
  - If two Gaussians are similar (i.e., two measures corroborate each other), the result becomes more certain

Kalman Gain

- Assume that:
  - x is the model prediction
  - z indicates the measurements
- The mean of the posterior is:

$$\mu = \frac{\sigma_x^2 \mu_z + \sigma_z^2 \mu_x}{\sigma_x^2 + \sigma_z^2} = \frac{\sigma_x^2}{\sigma_x^2 + \sigma_z^2}\mu_z + \frac{\sigma_z^2}{\sigma_x^2 + \sigma_z^2}\mu_x = K\mu_z + (1-K)\mu_x$$

- The Kalman Gain K:
  - Is the scaling term that mixes the prediction and the measurement
  - Depends on the ratio of uncertainty of prior and measurement

Kalman Pseudo-Algorithm

- The typical formulation of the Kalman filter is in terms of the "orthogonal projection" approach to minimize mean squared error
  - Instead of a Bayesian formulation
- Typical symbols used in Kalman literature:
  - x: state
  - P: variance of state (uncertainty, belief)
  - $f()$: system model
  - Q: system model error
  - z: measurement
  - R: measurement noise
- **Initialization**
  - Initialize state of filter $x = x_0$
  - Initialize belief in the state $P = P_0$
- **Predict**
  - Use system model to predict state at the next time step $x = f(x)$
  - Adjust belief to account for uncertainty in prediction $P = P + Q$
- **Update**
  - Get measurement z and belief about its accuracy R
  - Compute residual between estimated state x and z: $y = z - x$
  - Compute scaling factor (Kalman K) based on accuracy of prediction P and measurement R

- One Dimensional Kalman Filters
- ***Multivariate Gaussians***
- Multivariate Kalman Filters

Multivariate State

- Often the state variable is multivariate, e.g.,
  - Position and velocity of a dog (probably uncorrelated)
  - Height and weight of an adult (correlated)
- **Variance** is a measure of how a population varies, e.g.,
  - Variance $= 0$ means constant
  - Large variance means lots of variation
- **Covariance** are correlated variances
  - E.g., height and weight are generally positively correlated
- **Covariance matrix**
  - The diagonal contains the variance for each variable
  - The off-diagonal elements contain the covariance between i and j variables
  - The covariance matrix is symmetric
- Correlation allows prediction
  - E.g., "as winter comes you predict you will spend more on heating your house"

Multivariate Gaussian

- The marginal of a multivariate Gaussian is 1-d Gaussian
- Consider a contour plot (i.e., the intersection of a 2-d Gaussian $z = f(x, y)$ with a plane $z = c$)
  - The contour plot is always an ellipses

Multiplying Two Multivariate Gaussians

- Given two multivariate Gaussians $\sim Normal(\underline{\boldsymbol{\mu}}_i, \underline{\underline{\boldsymbol{\Sigma}}}_i)$
- The product of the Gaussians is still Gaussian $\sim Normal(\underline{\boldsymbol{\mu}}, \underline{\underline{\boldsymbol{\Sigma}}})$

$$\underline{\boldsymbol{\mu}} = \underline{\underline{\boldsymbol{\Sigma}}}_2(\underline{\underline{\boldsymbol{\Sigma}}}_1 + \underline{\underline{\boldsymbol{\Sigma}}}_2)^{-1}\underline{\boldsymbol{\mu}}_1 + \underline{\underline{\boldsymbol{\Sigma}}}_1(\underline{\underline{\boldsymbol{\Sigma}}}_1 + \underline{\underline{\boldsymbol{\Sigma}}}_2)^{-1}\underline{\boldsymbol{\mu}}_2$$
$$\underline{\underline{\boldsymbol{\Sigma}}} = \underline{\underline{\boldsymbol{\Sigma}}}_1(\underline{\underline{\boldsymbol{\Sigma}}}_1 + \underline{\underline{\boldsymbol{\Sigma}}}_2)^{-1}\underline{\underline{\boldsymbol{\Sigma}}}_2$$

- **Note**: this is a generalization of the 1-d case to multivariate

$$\mu = \frac{\mu_1\sigma_2^2 + \mu_2\sigma_1^2}{\sigma_1^2 + \sigma_2^2}$$
$$\sigma^2 = \frac{\sigma_1^2\sigma_2^2}{\sigma_1^2 + \sigma_2^2}$$

replacing:

- σ^2 with covariance matrix $\underline{\underline{\boldsymbol{\Sigma}}}$
- Division with matrix inversion

Multivariate Filtering

- Covariance structure helps improve the estimate, e.g.,
  - You know an airplane direction can't change quickly
  - Knowing an approximate value for the velocity helps constrain possible next positions
- **E.g., airplane**
  - You are tracking a plane moving in a direction (1-d problem)
  - At time 1, you are fairly certain about the position $x = 0$, but you don't know the velocity
    - You plot position and velocity on an x-y plane
    - The covariance matrix between position and velocity is narrow and tall
    - It is narrow on the x-axis since you know that the position is around $x = 0$
    - It is tall on the y-axis because of your lack of knowledge about velocity
  - After 1 sec, you get a position update of $x = 5$
    - You can infer that the velocity is 5/s
    - The covariance matrix is then stretched diagonally

- One Dimensional Kalman Filters
- Multivariate Gaussians
- ***Multivariate Kalman Filters***
    - Tracking a Dog with a Kalman Filter

Notation

- A Bayesian notation $a|b$ means "a given the evidence of b"
  - The prior is $\underline{\hat{\boldsymbol{x}}}_{t|t-1}$, since you know only the information at time $t-1$, i.e., the previous state
  - The posterior is $\underline{\hat{\boldsymbol{x}}}_{t|t}$, since you know all the information at time t, i.e., the measurement
- **A simpler notation:**
  - Indicate the "prior" version of the variables (i.e., after the system update) with an overline (E.g., $\overline{x}$, $\overline{\boldsymbol{x}}$, $\overline{\boldsymbol{X}}$)
  - Omit the indices $t+1$ and t and use an assignment notation (representing "update in place" of a variable):

$$x = x + 1$$

    instead of the mathematical notation using a different variable for each time step:

$$x_{t+1} = x_t + 1$$

  - With this notation:
    - The prior is $\overline{x} = \hat{x}_{t|t-1}$
    - The posterior is $x = \hat{x}_{t|t}$

SCIENCE ACADEMY

Multivariate Kalman Filter

- With the previous notation:
  - State update: $\overline{\boldsymbol{x}} = \boldsymbol{F}\boldsymbol{x} + \boldsymbol{B}\boldsymbol{u}$
  - State uncertainty: $\overline{\boldsymbol{P}} = \boldsymbol{F}\boldsymbol{P}\boldsymbol{F}^T + \boldsymbol{Q}$
  - Residual: $\boldsymbol{y} = \boldsymbol{z} - \boldsymbol{H}\overline{\boldsymbol{x}}$
  - Kalman gain: $\boldsymbol{K} = \overline{\boldsymbol{P}}\boldsymbol{H}^T(\boldsymbol{H}\overline{\boldsymbol{P}}\boldsymbol{H}^T + \boldsymbol{R})^{-1}$
  - Updated state: $\boldsymbol{x} = \overline{\boldsymbol{x}} + \boldsymbol{K}\boldsymbol{y}$
  - Update state uncertainty: $\boldsymbol{P} = (\boldsymbol{I} - \boldsymbol{K}\boldsymbol{H})\overline{\boldsymbol{P}}$
- Where
  - $\boldsymbol{x}$ and $\boldsymbol{P}$ are the state mean and covariance
  - $\boldsymbol{F}$ is the state transition function
  - $\boldsymbol{Q}$ is the system error (i.e., the noise in the model assessment)
  - $\boldsymbol{B}$ and $\boldsymbol{u}$ model the control inputs to the system
  - $\boldsymbol{H}$ is the measurement function
  - $\boldsymbol{z}$ and $\boldsymbol{R}$ are the measurement mean and covariance
  - $\boldsymbol{y}$ is the residual
  - $\boldsymbol{K}$ is the Kalman gain
- Use the system model to predict the next state
  - When we multiply $\boldsymbol{F}$ to $\boldsymbol{x}$ we get the prior (i.e., the state before seeing any measurement)
- Form an estimate between the prior and the measurement

From Univariate to Multivariate Kalman Filter

- Let's compare

Definition	Univariate (Bayesian)	Univariate (Kalman)	Multivariate (Kalman)
State update	$\overline{\mu} = \mu + \mu_f$	$\overline{x} = x + dx$	$\overline{\boldsymbol{x}} = \boldsymbol{F}\boldsymbol{x} + \boldsymbol{B}\boldsymbol{u}$
State uncertainty	$\overline{\sigma}^2 = \sigma^2 + \sigma_f^2$	$\overline{P} = P + Q$	$\overline{\boldsymbol{P}} = \boldsymbol{F}\boldsymbol{P}\boldsymbol{F}^T + \boldsymbol{Q}$
Residual		$y = z - \overline{x}$	$\boldsymbol{y} = \boldsymbol{z} - \boldsymbol{H}\overline{\boldsymbol{x}}$
Kalman gain		$K = \frac{\overline{P}}{\overline{P}+R}$	$\boldsymbol{K} = \overline{\boldsymbol{P}}\boldsymbol{H}^T(\boldsymbol{H}\overline{\boldsymbol{P}}\boldsymbol{H}^T + \boldsymbol{R})^{-1}$
Updated state	$\hat{\mu} = \frac{\overline{\sigma}^2\mu_z + \sigma_z^2\overline{\mu}}{\overline{\sigma}^2+\sigma_z^2}$	$x = \overline{x} + Ky$	$\boldsymbol{x} = \overline{\boldsymbol{x}} + \boldsymbol{K}\boldsymbol{y}$
Upd. state uncertainty	$\sigma^2 = \frac{\overline{\sigma}^2\sigma_z^2}{\overline{\sigma}^2+\sigma_z^2}$	$P = (1-K)\overline{P}$	$\boldsymbol{P} = (\boldsymbol{I} - \boldsymbol{K}\boldsymbol{H})\overline{\boldsymbol{P}}$

Designing a Kalman filter

- The designer of the model needs to design:
  - The form of the state $\underline{\boldsymbol{x}}$ and $\underline{\underline{\boldsymbol{P}}}$
  - The system model $\underline{\underline{\boldsymbol{F}}}$ and $\underline{\underline{\boldsymbol{Q}}}$
  - The measurement $\underline{\boldsymbol{z}}$ and $\underline{\underline{\boldsymbol{R}}}$
  - The measurement function $\underline{\underline{\boldsymbol{H}}}$
  - The control inputs $\underline{\underline{\boldsymbol{B}}}$ and $\underline{\boldsymbol{u}}$ if there are control inputs

- One Dimensional Kalman Filters
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- Multivariate Kalman Filters
    - ***Tracking a Dog with a Kalman Filter***

Tracking 1D Dog: Problem formulation

- There is a dog moving on a 1-d track
- The dog moves approximately 1 meter per step
  - The velocity has variance due to noise/imperfect model specification
- There is a sensor that measures the position of the dog
  - The sensor has a certain error
- Time is discrete

(Nuvolo)